



Gas analysis — Preparation of calibration gas mixtures using dynamic volumetric methods

Part 7: Thermal mass-flow controllers

(ISO 6145-7:2009)

Gasanalyse — Herstellung von Kalibriergasgemischen mit Hilfe von dynamisch-volumetrischen Verfahren — Teil 7: Thermische Massendurchflussregler
(ISO 6145-7:2009)

Analyse des gaz — Préparation des mélanges de gaz pour étalonnage à l'aide de méthodes volumétriques dynamiques — Partie 7: Régulateurs thermiques de débit massique
(ISO 6145-7:2009)

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National Foreword

Due to some misprints in the present ÖNORM EN ISO a new edition has been published without prior public enquiry. The English version of ÖNORM EN ISO 6145-7 will be withdrawn and published again with the following correction:

Figure 1 has been replaced.

English Version

**Gas analysis - Preparation of calibration gas mixtures using
dynamic volumetric methods - Part 7: Thermal mass-flow
controllers (ISO 6145-7:2009)**

Analyse des gaz - Préparation des mélanges de gaz pour
étalonnage à l'aide de méthodes volumétriques
dynamiques - Partie 7: Régulateurs thermiques de débit
massique (ISO 6145-7:2009)

Gasanalyse - Herstellung von Kalibriergasgemischen mit
Hilfe von dynamisch-volumetrischen Verfahren - Teil 7:
Thermische Massendurchflussregler (ISO 6145-7:2009)

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Foreword

The text of ISO 6145-7:2009 has been prepared by Technical Committee ISO/TC 158 “Analysis of gases” of the International Organization for Standardization (ISO) and has been taken over as EN ISO 6145-7:2010.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2011, and conflicting national standards shall be withdrawn at the latest by June 2011.

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Endorsement notice

The text of ISO 6145-7:2009 has been approved by CEN as a EN ISO 6145-7:2010 without any modification.

**Gas analysis — Preparation of calibration
gas mixtures using dynamic volumetric
methods —**

**Part 7:
Thermal mass-flow controllers**

*Analyse des gaz — Préparation des mélanges de gaz pour étalonnage
à l'aide de méthodes volumétriques dynamiques —*

Partie 7: Régulateurs thermiques de débit massique



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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of technical committees is to prepare International Standards. Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights.

ISO 6145-7 was prepared by Technical Committee ISO/TC 158, *Analysis of gases*.

This second edition cancels and replaces the first edition (ISO 6145-7:2001). In preparation of the first edition, it was assumed that each thermal mass-flow controller (TMC) would be configured for use at its optimum performance, and the uncertainty in the method was estimated on that basis. In this edition, therefore, extra precautionary text has been added to make it clear that the method shall not be employed, for example, to make a 10:1 binary mixture by using two thermal mass-flow controllers of identical range with one operated at its maximum, say, of 1 000 ml/min and the other at 100 ml/min. In the first edition, this necessary provision was only stated briefly in an informative annex; it has now been expanded and stated more explicitly in a normative part. Another major update is separation of the original Clause 3 into two clauses, one of which (Clause 4) defines the principle while the other (Clause 5) presents additional explanation to the user. The latter of these clauses now includes the necessary requirements. By introducing two new and relevant bibliographic references, the understanding of Annex B has been improved. Finally, some typing errors have been corrected.

ISO 6145 consists of the following parts, under the general title *Gas analysis — Preparation of calibration gas mixtures using dynamic volumetric methods*:

- *Part 1: Methods of calibration*
- *Part 2: Volumetric pumps*
- *Part 4: Continuous syringe injection method*
- *Part 5: Capillary calibration devices*
- *Part 6: Critical orifices*
- *Part 7: Thermal mass-flow controllers*
- *Part 8: Diffusion method*
- *Part 9: Saturation method*
- *Part 10: Permeation method*
- *Part 11: Electrochemical generation*

ISO 6145-3, entitled *Periodic injections into a flowing gas stream*, has been withdrawn.